PRE-PROCESSING

DESIGN SEARCH

CROSS EVAL. & SYNTHESIS

Input Data

Cape Town Telemetry

December 2025 Sebenza

Ranks, road network, CITP

Dwell Detection

Stop Events

Home Locations

Extract Route Library

Observed Routes

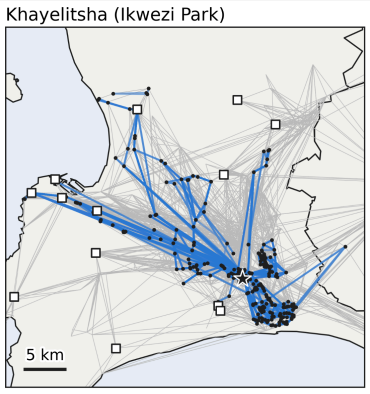
Reverse Routes

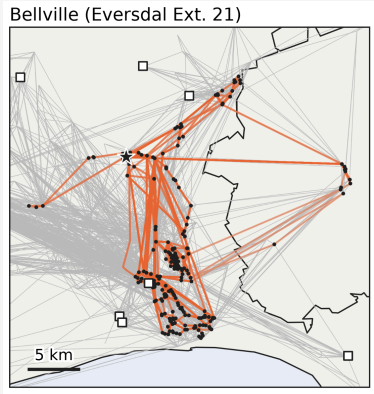
Detect Vehicle Communities

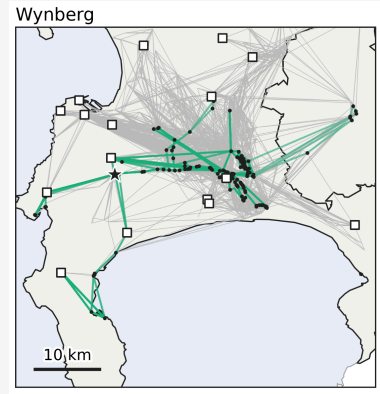
8 Communities

Apply screening

3 SCREENED COMMUNITIES







Data-replay Simulation M_R

- Observed movement is fixed
- Charging only at observed stops

Representation check $\mathbf{x}_S^\star$
Replay rules on simulated movement

Agent-based Simulation M_A

- Movement from driver decisions
- EVs compete with ICE taxis
- Calibrated to observed operation

Same vehicles, EV values, sites, and design space $\mathbf{x} \in \mathscr{X}$

Replay Design

Every design evaluated.

Selected design $\mathbf{x}_R^\star$

Cross-evaluation in M_A

ABM outcome $J_A(\mathbf{x}_R^\star)$

Compare

ABM outcome $J_A(\mathbf{x}_A^\star)$

ABM Design Search

Search and validation.

Selected design $\mathbf{x}_A^\star$

Latent influence

UNOBSERVED FACTORS

Unsampled minibus taxis

External passenger demand

Wider (unobserved) network